

The History of the Internet

From ARPANET to the Modern Web

1. Origins — ARPANET (1960s)

The Internet traces its roots to the late 1960s with the creation of ARPANET (Advanced Research Projects Agency Network), funded by the United States Department of Defense. The first message ever transmitted over ARPANET was sent on October 29, 1969 by UCLA student Charley Kline to the Stanford Research Institute. The message was supposed to be "LOGIN" but the system crashed after just the first two letters — "LO" — making it the first two characters ever sent over the internet.

ARPANET was designed with a key principle in mind: packet switching. Instead of sending data as a continuous stream, information is broken into small packets, each traveling independently across the network and reassembled at the destination. This made the network highly resilient — if one route failed, packets could take a different path.

2. TCP/IP — The Language of the Internet (1970s)

In 1974, Vint Cerf and Bob Kahn published a landmark paper introducing the Transmission Control Protocol (TCP), and later the Internet Protocol (IP). Together, TCP/IP became the foundational communication protocol of the internet and is still used today. Vint Cerf is often referred to as one of the "Fathers of the Internet" for this contribution.

On January 1, 1983, ARPANET officially switched to TCP/IP — a date now known as the "Flag Day" of the internet. This transition was the moment the modern internet was born.

3. The World Wide Web (1989–1991)

The internet and the World Wide Web are often used interchangeably, but they are not the same thing. The internet is the physical infrastructure — cables, routers, servers. The World Wide Web (WWW) is a service that runs on top of the internet.

The Web was invented by British scientist Tim Berners-Lee in 1989 while working at CERN in Switzerland. He proposed a system of hyperlinked documents to help scientists share information. In 1991, the first website in history went live at the address `info.cern.ch`. It described what the World Wide Web was and how one could use it.

Three core technologies underpin the Web: HTML (HyperText Markup Language) for structuring content, HTTP (HyperText Transfer Protocol) for transferring data, and URLs (Uniform Resource Locators) for identifying resources.

4. The Browser Wars (1990s)

The release of the Mosaic browser in 1993 by Marc Andreessen and Eric Bina at the National Center for Supercomputing Applications (NCSA) was transformative. It was the first browser to display images inline with text, making the Web accessible to everyday users.

Netscape Navigator launched in 1994 and quickly dominated the browser market. Microsoft responded with Internet Explorer in 1995, bundling it with Windows 95. The fierce competition between Netscape and Internet Explorer throughout the late 1990s is known as the First Browser War. Internet Explorer eventually won, capturing over 90% of market share by 2002, largely because it was free and pre-installed on Windows.

5. The Dot-Com Boom and Bust (1995–2001)

The late 1990s saw an explosion of internet-based companies funded by venture capital. Companies like Amazon (founded 1994), Google (founded 1998), and eBay (founded 1995) launched during this era. Many startups raised millions of dollars with little to no revenue, betting that internet traffic alone would eventually be profitable.

The NASDAQ index peaked in March 2000 and then collapsed, losing nearly 78% of its value by October 2002. Hundreds of dot-com companies went bankrupt. However, the companies that survived — Amazon, Google, eBay — went on to become some of the most valuable corporations in history.

6. Web 2.0 and Social Media (2004–2010)

The term "Web 2.0" was popularized by Tim O'Reilly in 2004 to describe a new generation of the Web characterized by user-generated content, interactivity, and collaboration. Unlike the static pages of Web 1.0, Web 2.0 platforms allowed users to create and share content.

Key milestones of this era include: Facebook launched in 2004 (originally limited to Harvard students), YouTube founded in 2005 and acquired by Google for \$1.65 billion in 2006, Twitter launched in 2006, and the Apple iPhone released in 2007, which made mobile internet browsing mainstream. The iPhone fundamentally changed how people accessed the internet — from desktop browsers to mobile apps.

7. The Cloud Era (2006–Present)

Cloud computing refers to delivering computing services — servers, storage, databases, networking, software — over the internet. Amazon Web Services (AWS) launched in 2006 and pioneered the modern cloud industry. Microsoft Azure followed in 2010, and Google Cloud Platform in 2011.

Today, an estimated 94% of enterprises use cloud services. The cloud enabled startups to launch without building physical servers, dramatically lowering the cost of starting an internet business. Services like Netflix, Spotify, and Airbnb are entirely cloud-dependent.

8. Internet Statistics Today

As of 2024, there are approximately 5.4 billion internet users worldwide, representing about 67% of the global population. Over 1.9 billion websites exist, though only a fraction are actively maintained. The average person spends approximately 6 hours and 37 minutes online per day.

Global internet traffic is measured in exabytes (1 exabyte = 1 billion gigabytes). Cisco estimates that internet traffic surpassed 400 exabytes per month in 2022. Video streaming accounts for over 65% of all downstream internet traffic, with Netflix alone responsible for approximately 15%.

The most visited website in the world is Google.com, followed by YouTube.com, Facebook.com, and Wikipedia.org.